[INCH-POUND]
A-A-52524/1
September 22, 1995
SUPERSEDING
MS51000F
27 October 1986

COMMERCIAL ITEM DESCRIPTION SPECIFICATION SHEET

SEAL, PLAIN ENCASED: SINGLE SEALING MEMBER, SPRING LOADED

The General Services Administration has authorized the use of this Commercial Item Description Specification Sheet (CIDSS) for all federal agencies.

The complete requirements for procuring the seals shall consist of this document and the issue in effect of A-A-52524.

1. SCOPE. This CIDSS covers seals to be used as rotary shaft seals to retain oil and other fluids.

2. SALIENT CHARACTERISTICS.

2.1 Materials.

2.1.1 Class 1 sealing member material. Class 1 sealing members shall be made of polymer-impregnated leather or chrome-vegetable-retanned leather which has been mildew resistant treated and suitable for normal operating temperatures (see A-A-52524).

2.1.2 Class 2 sealing member material. Class 2 sealing members shall be made of oil resistant synthetic elastomer suitable for normal operating temperature service (see A-A-52524).

2.1.3 Class 3 sealing member material. Class 3 sealing members shall be made of oil resistant elastomer suitable for extreme operating temperature service (see A-A-52524).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E, Warren, MI 48397-5000.

FSC 5330

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2.2 Design and construction. The seals (classes 1, 2, and 3 of CID A-A-52524) shall be designed and constructed in accordance with figure 1.

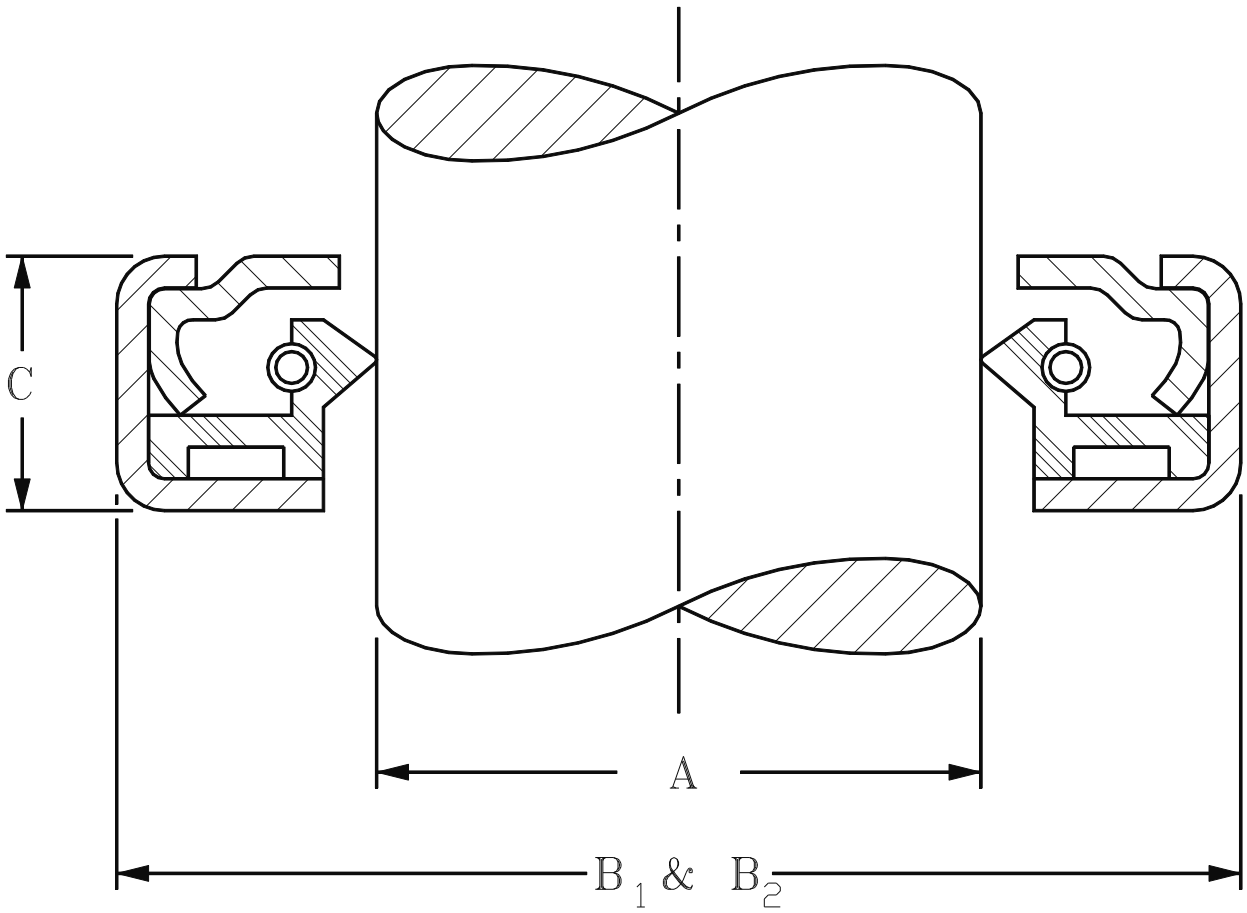

Figure 1. Seal design, construction, and part number.

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First Dash Number (Size)	A Shaft Dia (Nom)	B_1 Seal O.D. (1)	B_2 Housing Bore (Ref)	C Width (3)
101	.31	1.003	.999	.375
102	.38	1.003	.999	.375
103	.44	1.003	.999	.375
104	.50	1.254	1.250	.375
105	.56	1.254	1.250	.375
106	.63	1.379	1.375	.406
107	.69	1.379	1.375	.406
108	.75	1.379	1.375	.406
109	.81	1.379	1.375	.406
110	.88	1.379	1.375	.406
111	.94	1.629	1.625	.438
112	1.00	1.629	1.625	.438
113	1.06	1.832	1.828	.438
114	1.13	1.832	1.828	.438
115	1.19	1.988	1.984	.438
116	1.25	2.004	2.000	.438
117	1.31	2.066	2.062	.438
118	1.38	2.066	2.062	.438
119	1.44	2.378	2.374	.500
120	1.50	2.378	2.374	.500
121	1.56	2.506	2.502	.500
122	1.63	2.506	2.502	.500
123	1.69	2.754	2.750	.500
124	1.75	2.754	2.750	.500
125	1.81	2.754	2.750	.500
126	1.88	3.004	3.000	.500
127	1.94	3.004	3.000	.500
128	2.00	3.004	3.000	.500
129	2.06	3.004	3.000	.500
130	2.13	2.879	2.875	.438

First Dash Number (Size)	A Shaft Dia (Nom)	B_1 Seal O.D. (1)	B_2 Housing Bore (Ref)	C Width (3)
146	3.00	3.881	3.876	.500
147	3.06	4.130	4.125	.563
148	3.13	4.130	4.125	.563
149	3.19	4.255	4.250	.625
150	3.25	4.380	4.375	.625
151	3.31	4.380	4.375	.625
152	3.38	4.380	4.375	.625
153	3.44	4.380	4.375	.625
154	3.50	4.630	4.625	.625
155	3.56	4.630	4.625	.625
156	3.63	4.756	4.751	.625
157	3.69	4.756	4.751	.625
158	3.75	4.756	4.751	.625
159	3.81	4.756	4.751	.625
160	3.88	4.881	4.876	.625
161	3.94	4.881	4.876	.625
162	4.00	5.130	5.125	.625
163	4.06	5.255	5.250	.625
164	4.13	5.255	5.250	.625
165	4.19	5.255	5.250	.625
166	4.25	5.255	5.250	.625
167	4.31	5.506	5.501	.625
168	4.38	5.506	5.501	.625
169	4.44	5.506	5.501	.625
170	4.50	6.005	6.000	.625
171	4.56	6.005	6.000	.625
172	4.63	6.005	6.000	.625
173	4.69	6.005	6.000	.625
174	4.75	6.005	6.000	.625
175	4.81	6.005	6.000	.625

Figure 1. Seal design, construction, and part numbers - Continued.

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First Dash Number (Size)	A Shaft Dia (Nom)	B_1 Seal O.D. (1)	B_2 Housing Bore (Ref)	C Width (3)
131	2.13	2.879	2.875	.469
132	2.13	3.004	3.000	.500
133	2.19	3.255	3.250	.500
134	2.25	3.255	3.250	.500
135	2.31	3.255	3.250	.500
136	2.38	3.255	3.250	.500
137	2.44	3.255	3.250	.500
138	2.50	3.505	3.500	.500
139	2.56	3.755	3.750	.500
140	2.63	3.755	3.750	.500
141	2.69	3.755	3.750	.500
142	2.75	3.755	3.750	.500
143	2.81	3.755	3.750	.500
144	2.88	3.755	3.750	.500
145	2.94	3.755	3.750	.500

First Dash Number (Size)	A Shaft Dia (Nom)	B_1 Seal O.D. (1)	B_2 Housing Bore (Ref)	C Width (3)
176	4.88	6.005	6.000	.625
177	4.94	6.005	6.000	.625
178	5.00	6.256	6.250	.625
179	5.13	6.381	6.375	.625
180	5.25	6.506	6.500	.625
181	5.38	6.631	6.625	.625
182	5.50	6.756	6.750	.625
183	5.63	6.881	6.875	.625
184	5.75	7.006	7.000	.625
185	5.88	7.131	7.125	.625
186	6.00	7.506	7.500	.750
187	6.25	7.756	7.750	.750
188	6.50	8.006	8.000	.750
189	6.75	8.257	8.250	.750
190	7.00	8.507	8.500	.750

NOTES:

1. Dimensions are in inches. Tolerances are ± .002 for seal outside diameters (O.D.) up to and including 2.000 inches. Tolerances are +.003, -.002 for seal O.D. over 2.000 inches and up to and including 3.000 inches. Tolerances are + .004, -.002 for seal O.D. over 3.000 inches and up to and including 7.000 inches. Tolerances are +.005, -.002 for seal O.D. over 7.000 inches and up to and including 12.000 inches.
2. Maximum housing fillet radius .030 for seals up to and including 1.63 inch O.D. and .040 for seals over 1.63 inch O.D.
3. Width tolerance for seals shall be ± .015 inch for widths up to .400 and ± .020 inch for widths over .400 inch.
4. The part number consists of the CID designator, the CID number, the specification sheet number and 2 dash numbers (see paragraph 6.3 of A-A-52524). The first dash number indicates the size. The second dash number indicates the class. EXAMPLE: A52524/1-1-101-3 indicates size number 101 with class 3 sealing member.
5. The PIN dash numbers and the former MS part number dash numbers are the same. EXAMPLE: A-A-52524/1-101-3 cross references to MS51000-101-3.
6. The CIDSS is not intended to limit construction to features other than as shown herein by dimensions, notation, or referenced documents.

Figure 1. Seal design, construction, and part numbers - Continued.

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3. NOTES. (This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

3.1 Part or identification number (PIN). The PINs to be used for acquisition of seals are shown in paragraph 6.3 of A-A-52524.

3.2 Cross-reference data. Seals conforming to CIDSS are interchangeable/substitutable with seals conforming to MS51000F.

MILITARY INTERESTS:

Custodian:
Army - AT

Review activities:
Army - AR, AV, MI
DLA - IS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

PREPARING ACTIVITY:
Army - AT

(Project 5330-0896-01)

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